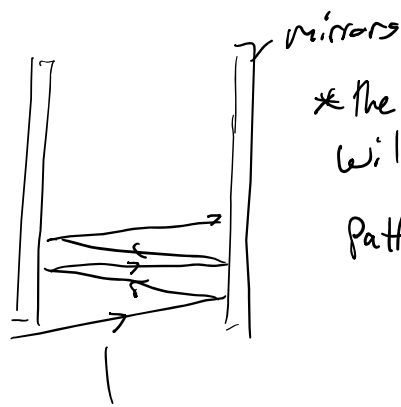


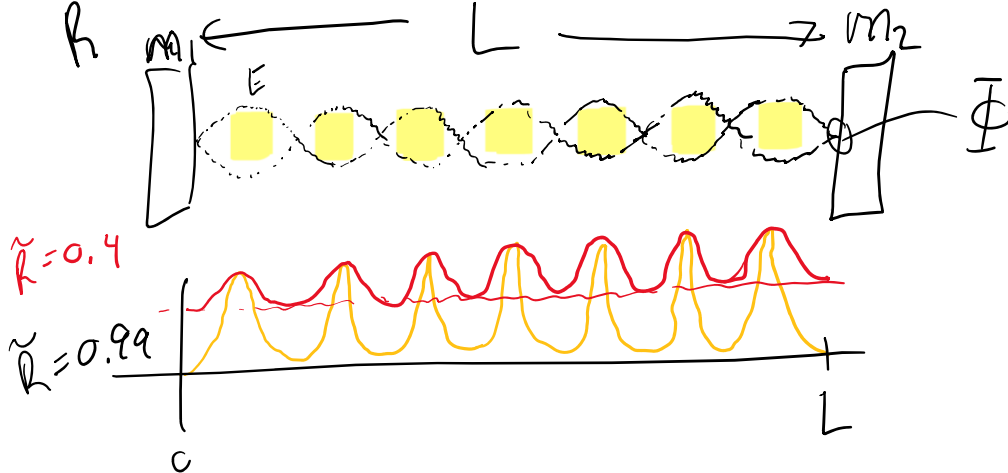
Lecture 16



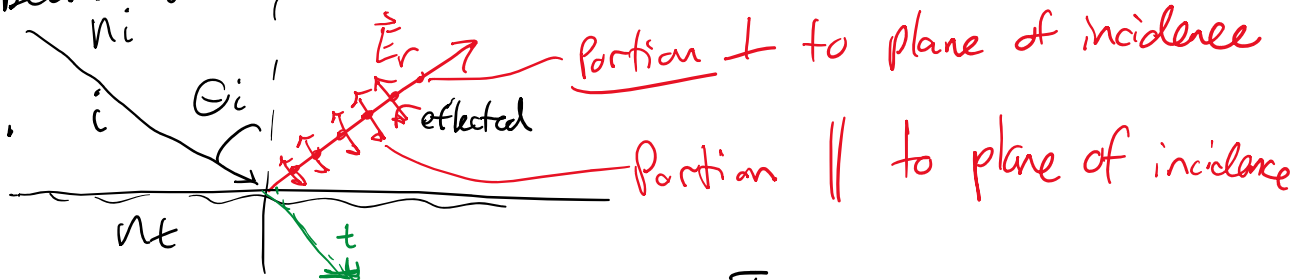
* the properties of how "shiny" the mirrors are will dictate the "goodness" of the interference patterns.

* while not drawn, the beams overlap as they bounce back and forth b/w the mirrors

* the measure of how "good" (Reflective) the mirrors are is the $\tilde{R}$



Let's look at how the incident beam angle affects the phase Φ b/w the incident beam and reflected beam for the $\perp$ component and $\parallel$ component of the reflected beam.

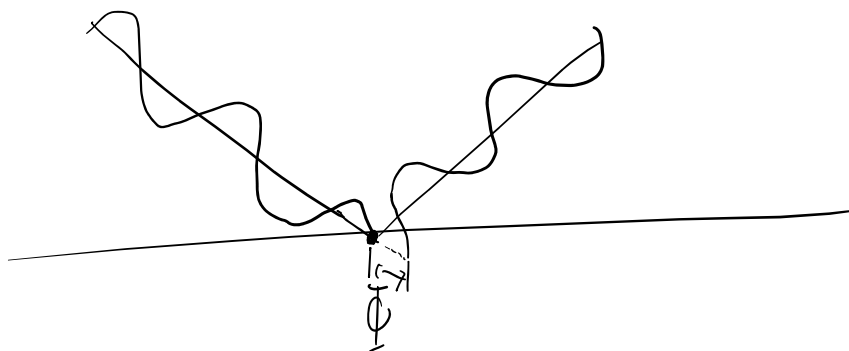


how does θ_i affect the spatial Φ b/w the E-field in the medium

$$\tilde{E}_{r\perp} \tan\left(\frac{1}{2}\Phi_{\perp}\right) = \frac{\left[\sin^2 \theta_i - (\tilde{n})^2\right]^{\frac{1}{2}}}{\cos \theta_i} \quad \tilde{n} = \frac{n_t}{n_i}$$

$\vec{E}_{\parallel}$

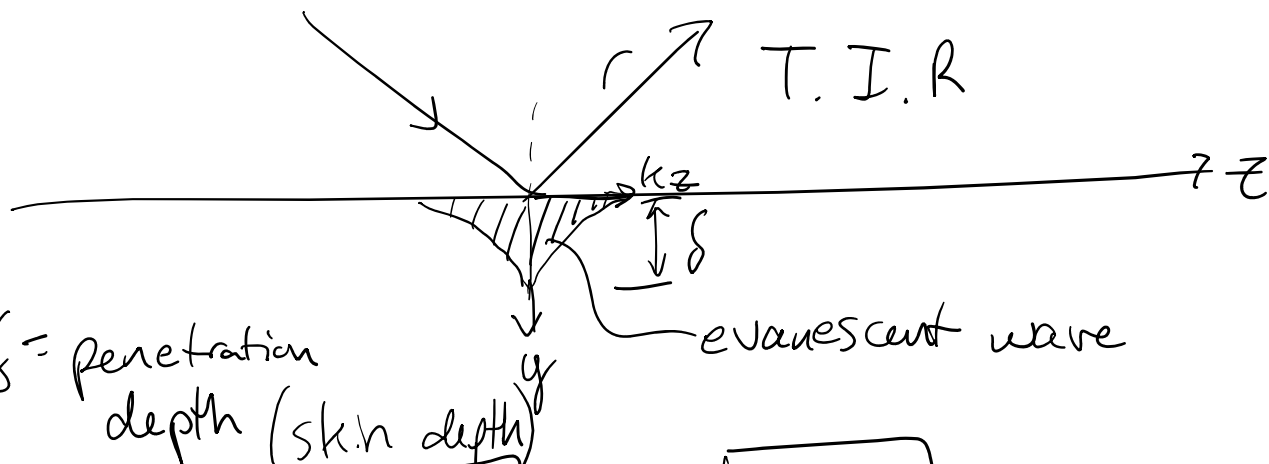
$$\tan\left(\frac{1}{2}\bar{\Phi}_{\parallel} + \frac{1}{2}\pi\right) = \frac{\left[\sin^2\theta_i - (\tilde{n})^2\right]^{\frac{1}{2}}}{(\tilde{n})^2 \cos(\theta_i)}$$



$n_t > n_i$ $\bar{\Phi}_{\perp}$ and $\bar{\Phi}_{\parallel}$ b/c imaginary

b/c = become

b/w = between



δ = penetration depth (skin depth)

$$\alpha(\text{attenuation}) = \frac{1}{\delta}$$

$\Rightarrow$

$$\delta = \frac{1}{\alpha}$$

$$k_z' = (k_i)(\sin\theta_i)$$

↑ propagation constant of the incident beam

$$q = \frac{1}{\delta} = \frac{2\pi n_t}{\lambda_0} \left[\left(\frac{n_i}{n_t} \right)^2 \sin^2 \theta_i - 1 \right]^{1/2}$$

↑
freespace wavelength

Reflectance ($\tilde{R}$) → intensity of reflected light with respect to the incident light

$$\tilde{R}_\perp = \frac{|\vec{E}_{r\perp}|^2}{|\vec{E}_{i\perp}|^2} = |r_\perp|^2$$

$$\tilde{R}_\parallel = \frac{|\vec{E}_{r\parallel}|^2}{|\vec{E}_{i\parallel}|^2} = |r_\parallel|^2$$

If complex

$$|E_{r\parallel}|^2 = (E_{r\parallel})(E_{r\parallel})^*$$

at normal incidence



$$\tilde{R} = \tilde{R}_\perp = \tilde{R}_\parallel = \left(\frac{n_i - n_t}{n_i + n_t} \right)^2$$

Transmittance (T): intensity of the transmitted light with respect to the incident light

$$T_{\perp} = \frac{n_t |E_{t\perp}|^2}{n_i |E_{i\perp}|^2} = \left(\frac{n_t}{n_i}\right) |t_{\perp}|^2$$

$$T_{\parallel} = \frac{n_t |E_{t\parallel}|^2}{n_i |E_{i\parallel}|^2} = \left(\frac{n_t}{n_i}\right) |t_{\parallel}|^2$$

Normal to n_t medium

$$T = T_{\parallel} = T_{\perp} = \frac{4n_i n_t}{(n_i + n_t)^2}$$

$$R + T = 1$$

@ critical angle (θ_c)

→ total internal Reflection

$$T_{\perp} = \frac{n_t (\cos \theta_t)}{n_i (\cos \theta_i)} |t_{\perp}|^2$$

$$T_{\parallel} = \frac{n_t (\cos \theta_t)}{n_i (\cos \theta_i)} |t_{\parallel}|^2$$

* Double check
at here.